

- 2 A long tube contains a liquid of density σ . Two rubber bands are placed on the tube towards the bottom as shown in Figure 2.1.

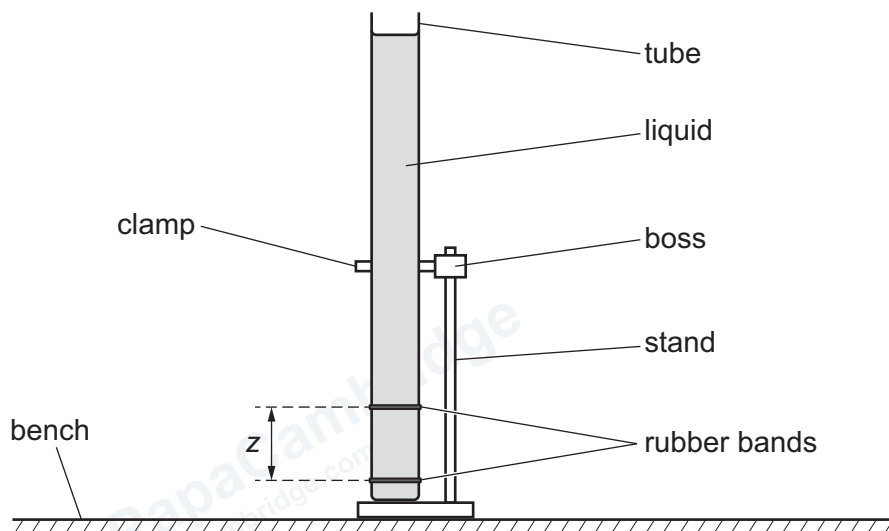


Figure 2.1

The distance between the two rubber bands is z .

A small steel ball of diameter d is dropped from the surface of the liquid. The time t for the ball to pass between the two rubber bands is measured.

The experiment is repeated with steel balls of different diameters.

It is suggested that t and d are related by the equation

$$\frac{z}{t} = \frac{gd^2(\rho - \sigma)}{18\beta}$$

where g is the acceleration of free fall, ρ is the density of steel and β is a constant.

- (a) A graph is plotted of t on the y -axis against $\frac{1}{d^2}$ on the x -axis.

Determine an expression for the gradient.

gradient = [1]

(b) Values of d and t are given in Table 2.1.

Table 2.1

d/mm	$\frac{1}{d^2}/\text{mm}^{-2}$	t/s
2.6 ± 0.1		6.10
3.0 ± 0.1		4.40
3.4 ± 0.1		3.35
4.1 ± 0.1		2.25
4.4 ± 0.1		1.95
5.0 ± 0.1		1.45

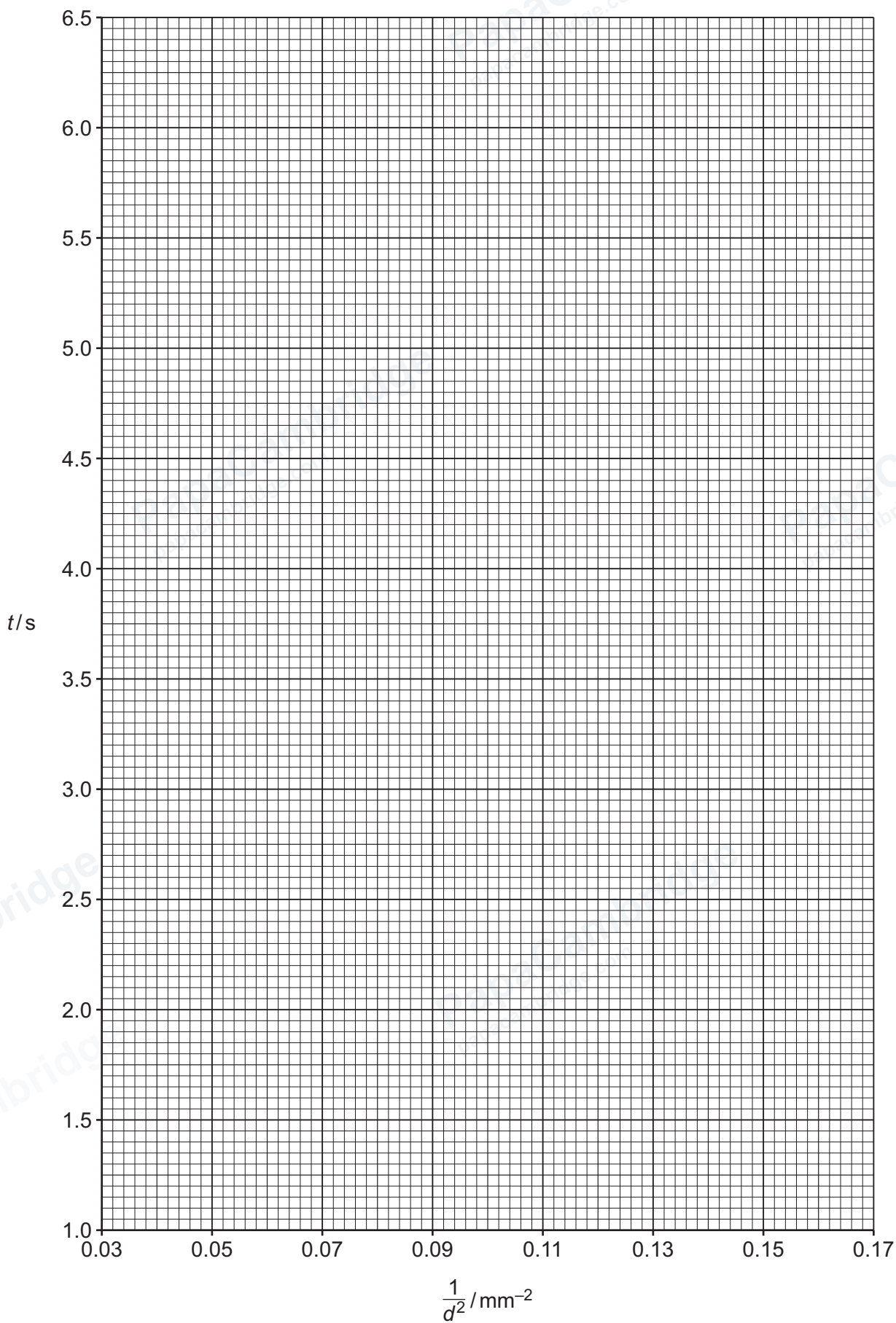
Calculate and record values of $\frac{1}{d^2}/\text{mm}^{-2}$ in Table 2.1. Include the absolute uncertainties in $\frac{1}{d^2}$. [2]

(c) (i) Plot a graph of t/s against $\frac{1}{d^2}/\text{mm}^{-2}$. Include error bars for $\frac{1}{d^2}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (d) The density ρ of steel is $(7850 \pm 50) \text{ kg m}^{-3}$.
The density σ of the liquid is $(1250 \pm 50) \text{ kg m}^{-3}$.

Calculate $(\rho - \sigma)$. Include the absolute uncertainty.

$$(\rho - \sigma) = \dots\dots\dots \text{ kg m}^{-3} \quad [1]$$

- (e) (i) Using your answers to **2(a)**, **2(c)(iii)** and **2(d)**, determine the value of β . Include an appropriate unit.

Data: $g = 9.81 \text{ m s}^{-2}$
 $z = (97 \pm 2) \text{ mm}$

$$\beta = \dots\dots\dots [2]$$

- (ii) Determine the percentage uncertainty in β .

$$\text{percentage uncertainty in } \beta = \dots\dots\dots \% [1]$$

- (f) The experiment is repeated using a steel ball with a diameter of $(1.20 \pm 0.01) \text{ cm}$. Determine the time t . Include the absolute uncertainty.

$$t = \dots\dots\dots \text{ s} [2]$$

[Total: 15]